Tech Bite #7: Reading Schematics, and What Circuits Do

How to read a schematic -- the standard symbols -- and what a handful of common circuits and components actually do.

Groups T6C + T6D · 23 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

The Big Picture

A **schematic** is a map of a circuit. This Bite teaches you to read one – to recognize the standard symbols – and covers what a handful of common circuits and components actually do. The exam shows you three diagrams and asks you to name numbered parts; the trick is to learn the symbols, not memorize the diagrams.

What a Schematic Is

A schematic is an electrical diagram drawn with **standard component symbols**. (T6C01) It accurately shows **how components are connected** to each other – not their physical sizes, their wire lengths, or what they look like. (T6C12)

The Three Exam Diagrams

The exam draws its symbol questions from these three figures. Study each symbol, then use the table to check yourself.

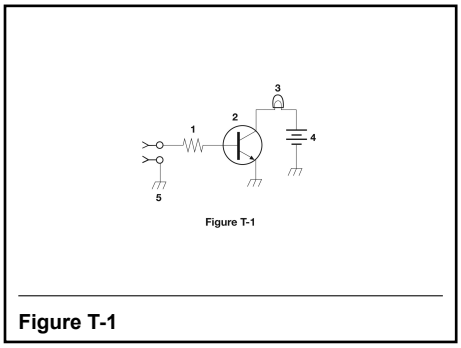


Figure T-1

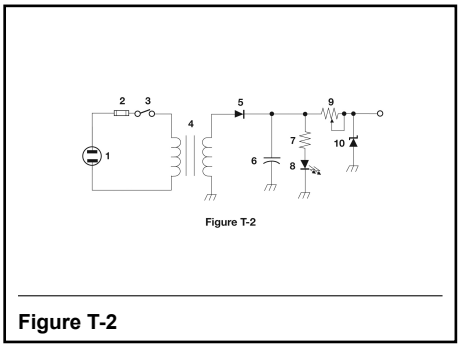


Figure T-2

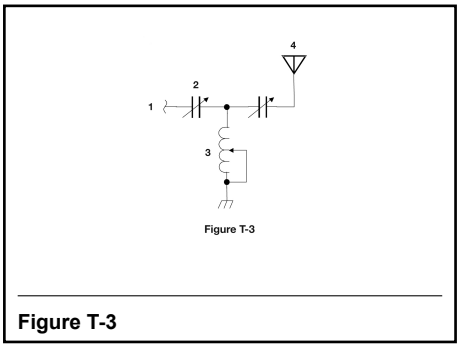


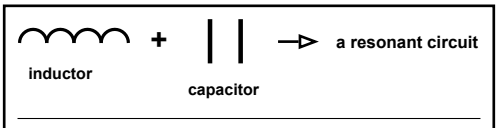
Figure T-3

Q	Where	What it is
T6C02	Figure T-1, part 1	**Resistor**
T6C03	Figure T-1, part 2	**Transistor** – its job is to control the flow of current (T6D10)
T6C04	Figure T-1, part 3	**Lamp**
T6C05	Figure T-1, part 4	**Battery**
T6C06	Figure T-2, part 6	**Capacitor**
T6C07	Figure T-2, part 8	**Light-emitting diode (LED)**
T6C08	Figure T-2, part 9	**Variable resistor**
T6C09	Figure T-2, part 4	**Transformer**
T6C10	Figure T-3, part 3	**Variable inductor**
T6C11	Figure T-3, part 4	**Antenna**

What Common Circuits and Components Do

Component	What it does	Q
Rectifier	Changes **AC into varying DC**.	T6D01
Transformer	Changes 120 V AC to a **lower (or higher) AC voltage**.	T6D06
Regulator	**Controls the voltage** coming from a power supply.	T6D05
Relay	An **electrically-controlled switch**.	T6D02
Meter	Displays an electrical quantity as a **numeric value**.	T6D04
LED	Commonly used as a **visual indicator**.	T6D07
Integrated circuit	Packs **several semiconductors and other parts into one package**.	T6D09
Shielded wire	**Prevents unwanted signals** from coupling into or out of the wire.	T6D03

One circuit worth knowing by name: a **resonant** (or **tuned**) circuit is simply an **inductor and a capacitor**, wired in series or in parallel. (T6D08, T6D11) Resonant circuits are how a radio tunes to one frequency and rejects the rest.



A resonant circuit is an inductor plus a capacitor.

Takeaway: A **schematic** uses standard symbols to show how parts **connect** – not their size or looks. Learn the symbols in figures T-1, T-2, and T-3 and the symbol questions are easy. As for the parts: a **rectifier** makes DC from AC, a **transformer** changes AC voltage, a **regulator** steadies voltage, a **relay** is a remote switch, and an **inductor plus a capacitor** makes a resonant circuit.

Bite 7 Question Bank (T6C + T6D)

T6C01. What is an electrical diagram using standard component symbols called?
A. Connection chart
B. Instrumentation system
C. Schematic
D. Flow chart

T6C02. What is component 1 in figure T-1?
A. Resistor
B. Transistor
C. Battery
D. Connector

T6C03. What is component 2 in figure T-1?
A. Resistor
B. Transistor
C. Indicator lamp
D. Connector

T6C04. What is component 3 in figure T-1?
A. Resistor
B. Transistor
C. Lamp
D. Ground symbol

T6C05. What is component 4 in figure T-1?
A. Resistor
B. Transistor
C. Ground symbol
D. Battery

T6C06. What is component 6 in figure T-2?
A. Resistor
B. Capacitor
C. Regulator IC
D. Transistor

T6C07. What is component 8 in figure T-2?
A. Resistor
B. Inductor
C. Regulator IC
D. Light emitting diode

T6C08. What is component 9 in figure T-2?
A. Variable capacitor
B. Variable inductor
C. Variable resistor
D. Variable transformer

T6C09. What is component 4 in figure T-2?
A. Variable inductor
B. Double-pole switch
C. Potentiometer
D. Transformer

T6C10. What is component 3 in figure T-3?
A. Connector
B. Meter
C. Variable capacitor
D. Variable inductor

T6C11. What is component 4 in figure T-3?
A. Antenna
B. Transmitter
C. Dummy load
D. Ground

T6C12. Which of the following is accurately represented in electrical schematics?
A. Wire lengths
B. Physical appearance of components
C. Component connections
D. All these choices are correct

T6D01. Which of the following devices or circuits changes an alternating current into a varying direct current signal?
A. Transformer
B. Rectifier
C. Amplifier
D. Reflector

T6D02. What is a relay?
A. An electrically-controlled switch
B. A current-controlled amplifier
C. An inverting amplifier
D. A pass transistor

T6D03. Which of the following is a reason to use shielded wire?
A. To decrease the resistance of DC power connections
B. To increase the current carrying capability of the wire
C. To prevent coupling of unwanted signals to or from the wire
D. To reduce receiver overload

T6D04. Which of the following displays an electrical quantity as a numeric value?
A. Potentiometer
B. Transistor
C. Meter
D. Relay

T6D05. What type of circuit controls the amount of voltage from a power supply?
A. Regulator
B. Oscillator
C. Filter
D. Phase inverter

T6D06. What component changes 120 V AC power to a lower AC voltage for other uses?
A. Variable capacitor
B. Transformer
C. Transistor
D. Diode

T6D07. Which of the following is commonly used as a visual indicator?
A. LED
B. FET
C. Zener diode
D. All these choices are correct

T6D08. Which of the following is combined with an inductor to make a resonant circuit?
A. Resistor
B. Zener diode
C. Potentiometer
D. Capacitor

T6D09. What is the name of a device that combines several semiconductors and other components into one package?
A. Transducer
B. Multi-pole relay
C. Integrated circuit
D. Transformer

T6D10. What is the function of component 2 in figure T-1?
A. Give off light when current flows through it
B. Supply electrical energy
C. Control the flow of current
D. Convert electrical energy into radio waves

T6D11. Which of the following is a resonant or tuned circuit?
A. An inductor and a capacitor in series or parallel
B. A linear voltage regulator
C. A resistor circuit used for reducing standing wave ratio
D. A circuit designed to provide high-fidelity audio

Answer Key

T6C01: C	T6C02: A	T6C03: B	T6C04: C	T6C05: D
T6C06: B	T6C07: D	T6C08: C	T6C09: D	T6C10: D
T6C11: A	T6C12: C	T6D01: B	T6D02: A	T6D03: C
T6D04: C	T6D05: A	T6D06: B	T6D07: A	T6D08: D
T6D09: C	T6D10: C	T6D11: A		